

Rod, thread and bead

X17 (2017) — English translation

A rod of mass M and length L is suspended by its ends on long identical threads fixed at point O . The threads are threaded through a small bead of mass m , which can move along them without friction (see figure). The acceleration due to gravity is g .

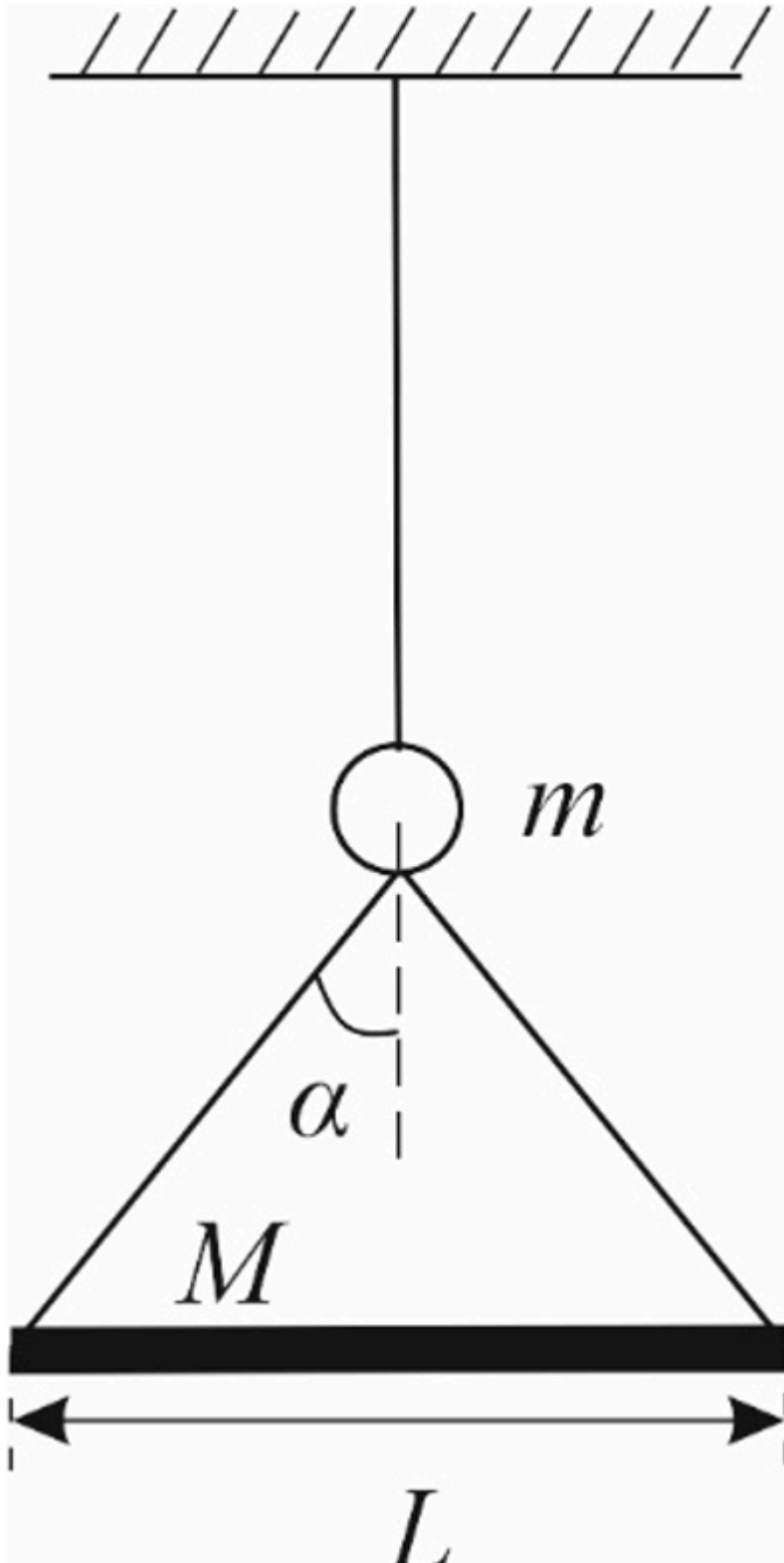


Fig. 1.

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|-----------|---|-------------|
| A1 | Determine the angle α_0 between the thread and the vertical for the case when the system is in equilibrium. | 1.40 |
| A2 | The bead is moved a short distance down and released. Find the period of small oscillations of the system around the equilibrium position (the movement of the bead and the rod occurs along a vertical straight line). | 5.60 |

Source: <https://pho.rs/p/301>